Task 3: Work with YourTTS

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Abstract

YourTTS (English, French and Brazilian Portuguese only) synthesized an English and a Lithuanian sentence in my voice without training. In both languages it copied my pitch well (129 Hz vs 123–126 Hz) but spoke about 12 % slower, with flatter intonation and no energy above 8 kHz. The English result sounded clear and like me; the Lithuanian one sounded unnatural. A dataset of 437 recordings (22.7 min) of EMO speaker M4, emotion neutral, was prepared for fine-tuning.

1. Model and supported languages

YourTTS is a multilingual text-to-speech model that can copy a new voice from a few seconds of speech without training (**zero-shot voice cloning**). It was loaded with the **Coqui TTS** library. `print(tts.languages)` returned `['en', 'fr-fr', 'pt-br']`: English, French and Brazilian Portuguese. **Lithuanian is not supported.**

2. Recordings and analysis method

Reference recordings. I recorded on an iPhone the English *"This is a test of the speech synthesis system."* (`reference_en.wav`) and the Lithuanian *"Tai yra šnekos sintezės sistemos testas."* (`reference_lt.wav`). Each served both as the **natural speech** sample and as the **voice reference**. The click at the end of each file (the tap that stopped the recording) was cut off and the stereo files were converted to mono (Table 1).

Recording	Duration	Sampling rate	Format	Channels
reference_en.wav (natural)	3.50 s	48 kHz	WAV, 16-bit PCM	1 (mono)
output_en.wav (synthesized)	3.84 s	16 kHz	WAV, 16-bit PCM	1 (mono)
reference_lt.wav (natural)	3.60 s	48 kHz	WAV, 16-bit PCM	1 (mono)
output_lt.wav (synthesized)	4.14 s	16 kHz	WAV, 16-bit PCM	1 (mono)

Table 1. Parameters of the natural and synthesized recordings.

Analysis. Each signal was loaded with librosa at its original sampling rate. The **waveform** shows amplitude over time, the **magnitude spectrum** (FFT of the whole signal, in dB) shows the energy at each frequency, and the **F0 contour** (pYIN, 60–400 Hz) shows the pitch. I also measured the **speech duration** without silence and the **share of energy at 4–8 kHz**.

3. English zero-shot synthesis

Procedure. The English sentence was synthesized with `tts.tts_to_file(text=..., speaker_wav="reference_en.wav", language="en")`, which produced `output_en.wav` in my voice.

Quality. The synthesized English sentence sounds **clear**, and the voice **sounds like me**.

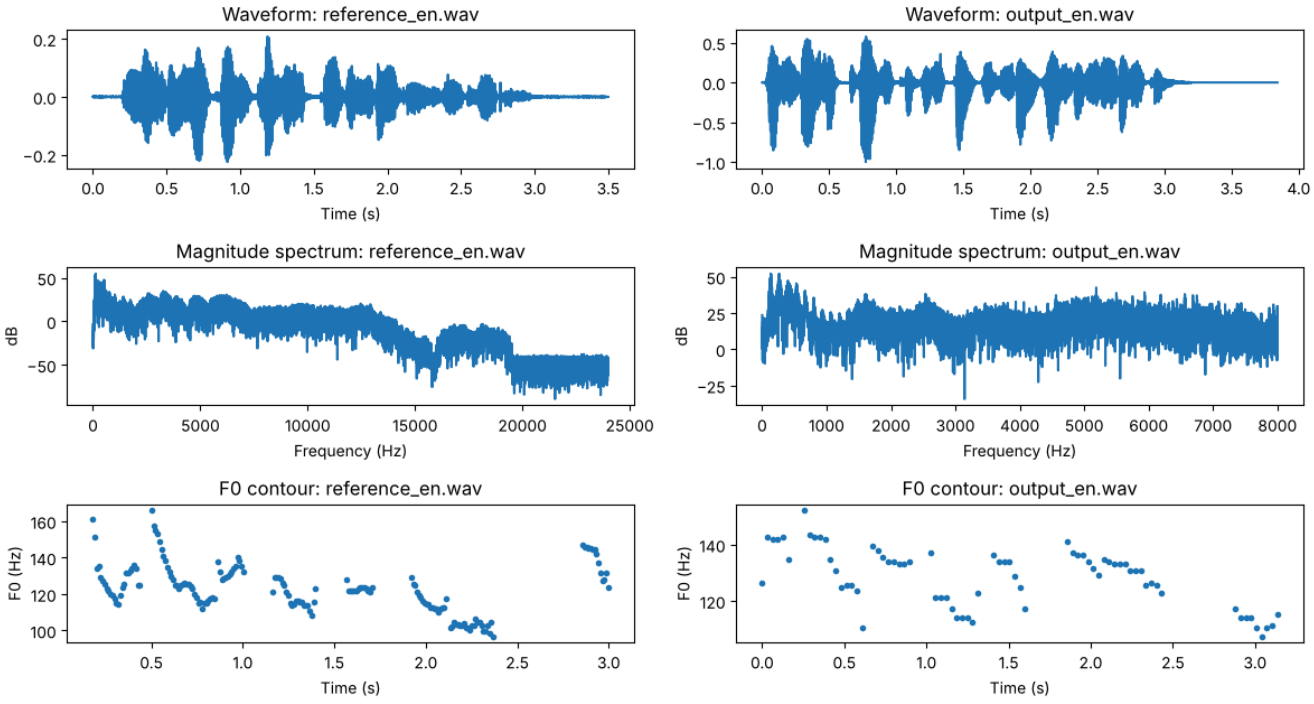


Figure 1. English: waveform, magnitude spectrum and F0 contour of natural (left) and synthesized (right) speech.

	EN natural	EN synthesized	LT natural	LT synthesized
Speech duration (silence removed)	2.83 s	3.17 s	3.10 s	3.46 s
Highest frequency	~13 kHz	8 kHz	~13 kHz	8 kHz
Energy at 4–8 kHz	5.9 %	18.7 %	4.8 %	7.3 %
Mean F0	123 Hz	129 Hz	126 Hz*	129 Hz

	EN natural	EN synthesized	LT natural	LT synthesized
F0 range	97–166 Hz	108–152 Hz	107–149 Hz*	108–163 Hz

*Table 2. Comparison of natural and synthesized speech. * Without F0 errors in the silence after the sentence; the script prints 117 Hz and 60–149 Hz.*

Duration. The synthesized speech is about **12 % slower** (3.17 s vs 2.83 s without silence) and starts with no silence before it.

Spectrum. YourTTS outputs **16 kHz** audio, so its spectrum stops at **8 kHz**, while my 48 kHz recording has speech energy up to about 13 kHz. The synthesized speech also has **much more energy at 4–8 kHz** (18.7 % vs 5.9 %), even during vowels, which is probably noise from the model's vocoder.

F0. The mean F0 is **129 Hz against my 123 Hz** (about one semitone higher), so my pitch was copied well. The range is smaller (108–152 vs 97–166 Hz), so the intonation is **flatter**, and the pitch peak on my last word (about 147 Hz) is missing.

4. Lithuanian zero-shot synthesis

Procedure. Lithuanian is not supported, so the sentence was synthesized with the **English setting** (`language="en"`), giving `output_lt.wav` .

Quality. The synthesized Lithuanian sentence **sounds weird**, not like natural Lithuanian, because the model reads it with English pronunciation rules.

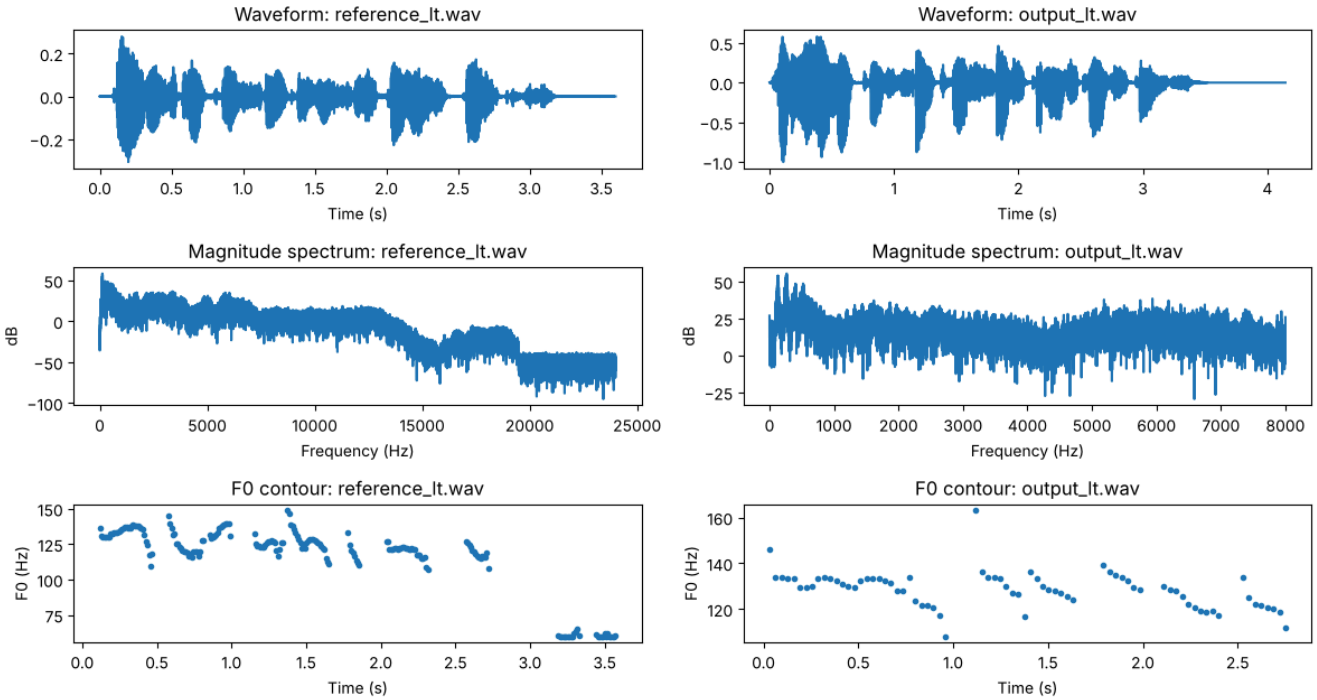


Figure 2. Lithuanian: waveform, magnitude spectrum and F0 contour of natural (left) and synthesized (right) speech.

Duration and spectrum. The synthesized speech lasts 3.46 s against 3.10 s (Table 2), again about **12 % slower**, so the model kept its own speaking rate and did not skip large parts of the text. As in English, the spectrum stops at 8 kHz; the extra energy at 4–8 kHz is smaller here (7.3 % vs 4.8 %).

F0. The mean F0 is close to mine (129 Hz vs 126 Hz); the 60 Hz points after 3.2 s in my plot are pYIN errors in the silence. The synthesized contour repeats the same fall on almost every word (about 135 → 120 Hz), while mine is more varied, with rises at 0.6–1.0 s and a peak of 148 Hz at 1.4 s.

5. Lithuanian dataset preparation

The dataset was built from the **Lithuanian Emotional Speech Corpus (EMO)** (Vilnius University, CLARIN-LT, https://hdl.handle.net/20.500.11821/100). Its metadata lists **349,696** transcribed recordings by five speakers (F2, F3, F4 female; M3, M4 male) in five emotions: D – joy, L – sadness, N – surprise, P – anger and Z – neutral. Only **one archive part**, `emo-wavs.zip.021` (312 MB), was downloaded.

Selection. The archive part is one piece of a split zip file, so a normal unzip program cannot open it; a short script (`extract_zip_part.py`) recovered the **437** complete WAV files inside it. Speaker and emotion are coded in each file name (e.g. `M4Z` = speaker M4, emotion Z), and all 437 files belong to speaker **M4** (male) and emotion **Z – neutral**, so all of them were selected. Their transcriptions were taken from `metadata_of_EMO_dataset.csv`.

Split and format. The files were renamed `0001.wav`, `0002.wav`, …, shuffled (seed 42) and split into **90 % training (394)** and **10 % validation (43)**. The metadata files use the Coqui *ljspeech* format, one line per recording: `0001|transcription|transcription`.

Parameter	Value	Parameter	Value
Speaker	M4 (male)	Recordings	437
Emotion	Z – neutral	Total duration	22.7 min
Sampling rate	96 kHz	Minimum duration	0.87 s
Bit depth	24 bit	Maximum duration	7.89 s
Channels	1 (mono)	Average duration	3.12 s
Training set	394	Validation set	43

Table 3. Selected EMO recordings.

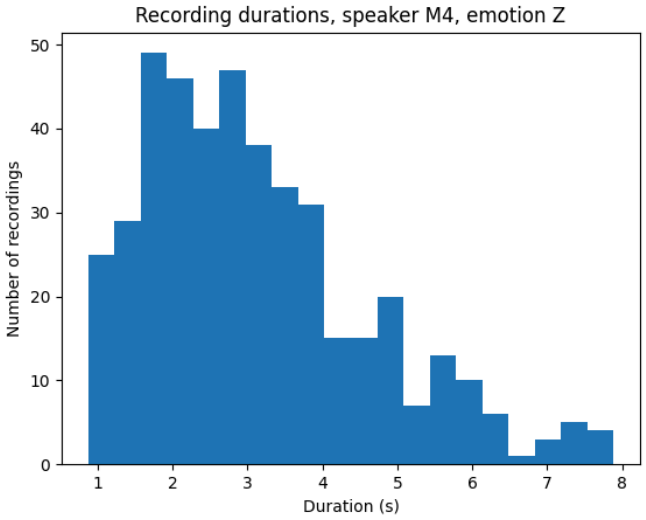


Figure 3. Histogram of recording durations for speaker M4, emotion Z.

Interpretation. Most recordings are short phrases of 1.5–4 s, with the peak around 2 s; fewer last more than 5 s, and only a handful reach 7–8 s. The audio is 96 kHz, 24-bit, much higher quality than the 16 kHz that YourTTS uses, so it would have to be resampled to 16 kHz before fine-tuning.

6. Conclusions

1. **Languages:** YourTTS supports English, French and Brazilian Portuguese, not Lithuanian.
2. **Voice:** in both languages the synthesized pitch was close to mine (129 Hz vs 123–126 Hz), but the speech was about 12 % slower, with flatter intonation, nothing above 8 kHz and extra noise at 4–8 kHz.
3. **Pronunciation:** the English sentence sounded clear and like me, but the Lithuanian one sounded weird because it was read with English pronunciation, so zero-shot synthesis is not enough for an unsupported language.
4. **Dataset:** 437 recordings (22.7 min) of speaker M4, emotion Z (neutral), were prepared from the EMO corpus: 394 for training and 43 for validation.